
```

1. """Three-Card Monte, by Al Sweigart al@inventwithpython.com
2. Find the Queen of Hearts after cards have been swapped around.
3. (In the real-life version, the scammer palms the Queen of Hearts so you
4. always lose.)
5. More info at https://en.wikipedia.org/wiki/Three-card\_Monte
6. View this code at https://nostarch.com/big-book-small-python-projects
7. Tags: large, card game, game"""
8.
9. import random, time
10.
11. # Set up the constants:
12. NUM_SWAPS = 16 # (!) Try changing this to 30 or 100.
13. DELAY      = 0.8 # (!) Try changing this 2.0 or 0.0.
14.
15. # The card suit characters:
16. HEARTS = chr(9829) # Character 9829 is '♥'
17. DIAMONDS = chr(9830) # Character 9830 is '♦'
18. SPADES = chr(9824) # Character 9824 is '♠'
19. CLUBS = chr(9827) # Character 9827 is '♣'
20. # A list of chr() codes is at https://inventwithpython.com/chr
21.
22. # The indexes of a 3-card list:
23. LEFT = 0
24. MIDDLE = 1
25. RIGHT = 2
26.
27.
28. def displayCards(cards):
29.     """Display the cards in "cards", which is a list of (rank, suit)
30.     tuples."""
31.     rows = ['', ' ', ' ', ' ', ' '] # Stores the text to display.
32.
33.     for i, card in enumerate(cards):
34.         rank, suit = card # The card is a tuple data structure.
35.         rows[0] += '   ' # Print the top line of the card.
36.         rows[1] += '|{}| '.format(rank.ljust(2))
37.         rows[2] += '| {} | '.format(suit)
38.         rows[3] += '|_{}| '.format(rank.rjust(2, '_'))
39.
40.
41.     # Print each row on the screen:
42.     for i in range(5):
43.         print(rows[i])
44.
45.
46. def getRandomCard():
47.     """Returns a random card that is NOT the Queen of Hearts."""
48.     while True: # Make cards until you get a non-Queen of Hearts.
49.         rank = random.choice(list('23456789JQKA') + ['10'])
50.         suit = random.choice([HEARTS, DIAMONDS, SPADES, CLUBS])
51.
52.         # Return the card as long as it's not the Queen of Hearts:
53.         if rank != 'Q' and suit != HEARTS:
54.             return (rank, suit)

```